High Voltage Technology: Laboratory Introduction

EEN442 Erasmus lab week at Hochschule Darmstadt (h_da) · Wednesday 21 October 2026, 10:30 to 13:00

Overview

This laboratory course focuses on **breakdown strength in air**, examining the electrical breakdown phenomena under different electrode configurations and voltage types. The session combines demonstrations by the lab staff with a partial discharge experiment, designed to provide practical understanding of high voltage engineering principles that are essential for electrical engineers working in power systems, insulation design, and high voltage equipment development[1][2].

Fundamental Concepts

Electrical Field Theory

High voltage technology is fundamentally governed by electrical field theory and Maxwell's equations. The behavior of electrical fields determines the dimensioning parameters for high voltage equipment. Key concepts include[1]:

- **Field strength calculation** using differential methods and superposition principles
- **Homogeneous vs. inhomogeneous electrical fields** and their impact on breakdown behavior
- **Geometry factors** that influence field distribution and maximum field strength

Voltage Types in Power Systems

The laboratory introduces students to various voltage types encountered in real power networks[1]:

- **AC voltages** (50/60 Hz) for power transmission
- **DC voltages** for HVDC transmission systems
- **Lightning impulse voltages** (1.2/50 µs) representing atmospheric disturbances
- **Switching impulse voltages** critical for systems above 300 kV

Insulation Materials and Breakdown Mechanisms

Understanding breakdown behavior in different insulating materials is crucial[1]:

- **Gaseous insulation** (air, SF_6): Townsend mechanism, streamer mechanism, leader mechanism
- **Discharge processes**: Corona discharge, partial discharge, complete breakdown
- **Polarity effects** in DC applications affecting breakdown voltage levels

Laboratory Programme

Demonstration 1A: AC Breakdown Testing

The lab staff demonstrate AC breakdown with different electrode geometries[2]:

- **Plate-plate configuration**: Uniform field distribution
- **Point-plate configuration**: Non-uniform field with field enhancement

Quantities discussed:

- Breakdown voltage (Ûd) as function of electrode spacing
- Average field strength (E_{mittel})
- Geometry factor ($p = (s + r) / r$)
- Utilization factor (η)
- Maximum field strength (E_{max})

Demonstration 1B: Pre-breakdown Phenomena

Demonstration of progressive breakdown stages in point-point configuration[2]:

- **Glow onset voltage** (Glimmeinsetzspannung)
- **Streamer onset voltage** (Büscheleinsetzspannung)
- **Spark onset voltage** (Funkeneinsetzspannung)

Demonstration: Lightning Impulse

A multi-stage impulse voltage generator applies 1.2/50 µs lightning impulses to a railway mast disconnector. See the lightning impulse handout.

Demonstration: Tesla Transformer

A spark gap and two tuned resonant circuits produce a high-frequency high voltage. See the Tesla and partial discharge handout.

Experiment: Partial Discharges in Solids

Partial discharges from a needle electrode in solid insulation, observed on the sine wave before breakdown. See the Tesla and partial discharge handout.

Safety Considerations

High voltage laboratory work requires strict safety protocols[2]:

- Mandatory safety briefing before experiment commencement
- Familiarization with grounding equipment
- Written confirmation of safety instruction understanding
- Proper use of earthing switches and protective equipment

Practical Applications

The laboratory concepts directly relate to real-world high voltage engineering[1]:

- **Power transmission systems**: Understanding insulation coordination for 420 kV and higher voltage levels
- **Lightning protection**: Design of protection systems based on impulse voltage withstand capability
- **Equipment testing**: AC routine tests and lightning impulse type tests per IEC standards
- **Insulation design**: Dimensioning rules for high voltage components

Learning Objectives

Upon completion, students will understand[1]:

- Basic dimensioning rules for HV systems and practical limitations
- Relationship between theoretical calculations and practical design constraints
- Development processes for high voltage equipment including type testing requirements
- Theoretical and practical sequences for high voltage testing procedures

This laboratory experience provides essential hands-on knowledge that bridges theoretical understanding with practical high voltage engineering applications, preparing students for careers in power systems, equipment manufacturing, and electrical safety engineering.

References

[1] T. Betz, High Voltage Technology and Electrical Fields, lecture notes, h_da. Available from the lab staff.

[2] T. Betz, Breakdown strength in air, lab instructions (German), h_da. Available from the lab staff.